

Assignment 3

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Code Implementation:

-The primary algorithm prepared for by this code is Stereo Correspondence It attempts to find the same point in two different images

-The process works by shifting a small window (block) from the left image across the corresponding row in the right image to find the best match based on pixel intensity.

-The fundamental calculation performed at the pixel level is Disparity which represents the horizontal displacement of an object between the two views:

$$d = x_{\text{left}} - x_{\text{right}}$$

Where x is the horizontal coordinate of a feature in the respective image planes.

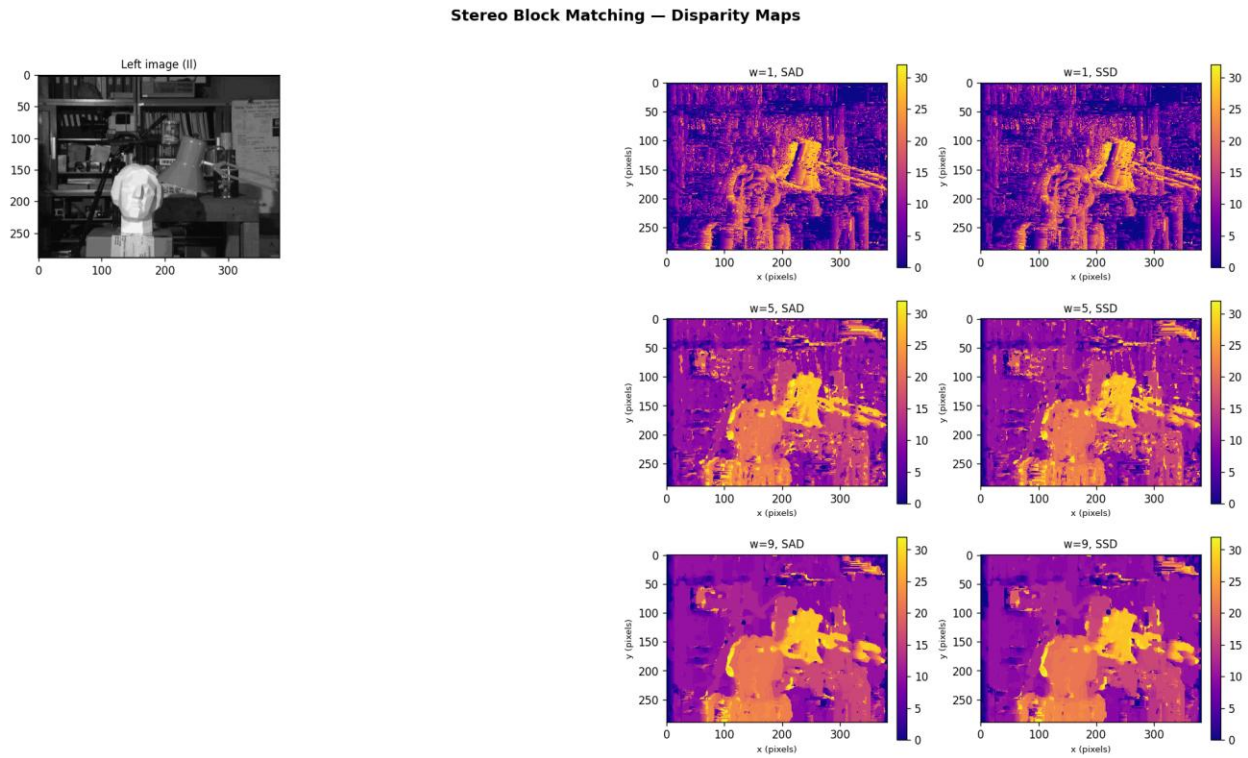
-Once the disparity is calculated, the system can derive physical distance Z from the camera This is based on the inverse relationship between shift and distance:

$$Z = (f * B) / d$$

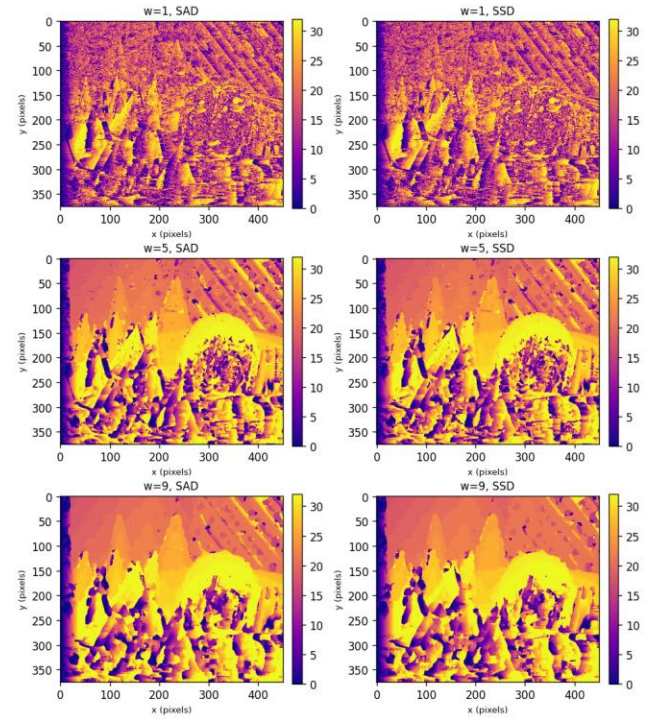
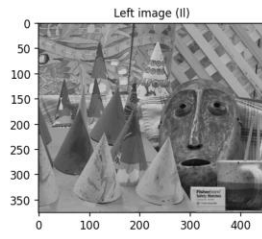
- f (Focal Length): The distance from the camera lens to the sensor.

- B (Baseline): The physical distance between the two camera centers.
- d (Disparity): The value derived from the matching algorithm.

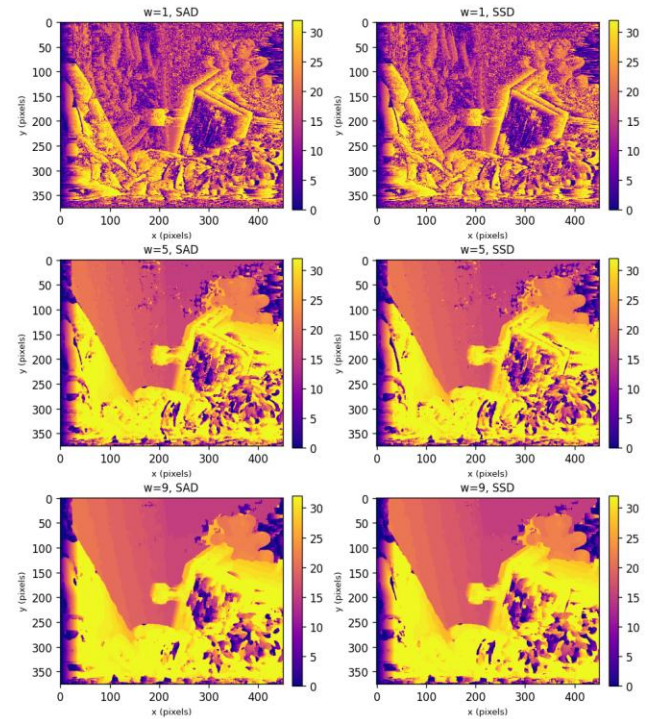
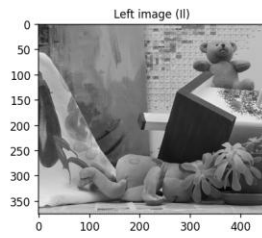
Test Runs:



Stereo Block Matching — Disparity Maps



Stereo Block Matching — Disparity Maps



DP

The DP algorithm treats stereo matching as a global optimization problem for each horizontal scanline. It builds a cost matrix, $\mathbf{D}$, where each cell represents the minimum cumulative cost to align pixels from the left and right images. This cost is calculated by combining normalized squared intensity differences with a skip penalty, c_0 , for occlusions. Finally, the algorithm backtracks from the bottom-right of the matrix to the top-left to determine the optimal matches and skips for the row.

Advantages

- **Global Optimization:** Finds the optimal alignment for an entire scanline simultaneously rather than making greedy, local decisions.
- **Edge Handling:** Explicitly accounts for occlusions via the "skip" penalty, leading to sharper object boundaries.
- **Noise Robustness:** Recursive calculation allows the algorithm to find logical paths even when individual pixel intensities are noisy.
- **Enforced Order:** Assumes the relative horizontal order of points is preserved, which simplifies the search space and stabilizes results.

Disadvantages

Disadvantages

- **Horizontal Streaking:** Since rows are processed independently, the lack of vertical constraints often creates horizontal artifacts.
- **Computational Complexity:** Filling an $N \times N$ matrix for every scanline is significantly slower than local block matching.
- **Ordering Violations:** Can fail in complex scenes where thin objects or sharp depth changes cause the physical ordering of points to shift.
- **Memory Intensive:** Requires storing a full cost matrix for each scanline during the calculation phase.

Output

